

**THE HERITAGE SCHOOL**  
**FIRST TERM EXAMINATION, 2026-27**  
**MATHEMATICS · CLASS 12 · PRACTICE PAPER 2**

**Maximum Marks: 80**

**Time Allotted: Three Hours**

**Instructions to Candidates**

- You are allowed an additional 15 minutes for only reading the paper.
- You must NOT start writing during reading time.
- The Question Paper consists of **twenty questions** and **four sections**: Sections A, B, C and D. **All questions are compulsory.**
- **Section A** consists of very short answer questions of **1 mark** each.
- **Section B** consists of short answer questions of **2 marks** each.
- **Section C** consists of moderately long answer questions of **3 marks** each.
- **Section D** consists of long answer questions of **5 marks** each.
- Internal choices have been provided in **three questions** in each of Sections B, C and D.
- While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
- The intended marks for questions or parts of questions are given in brackets [ ].
- All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer.
- Mathematical tables and graph papers are provided.

*Syllabus covered:* Relations and Functions; Inverse Trigonometric Functions; Matrices; Determinants; Continuity and Differentiability; Applications of Derivatives; Integrals; Application of Integrals; Linear Programming.

*Not examined in this paper (held back to the Pre-Board):* Differential Equations, Vectors, Three-dimensional Geometry, Probability.

**SECTION A – 20 MARKS**

**Question 1 [20 × 1]**

In sub-parts (i) to (xx), choose the correct option or answer the question as instructed.

(i) The relation  $R$  on  $\mathbb{R}$  defined by  $R = \{(a, b) : a \leq b^2\}$  is:

- (a) not reflexive
- (b) reflexive and symmetric
- (c) an equivalence relation
- (d) reflexive but not symmetric

(ii) The function  $f : \mathbb{R} \rightarrow \mathbb{R}$  given by  $f(x) = x^3$  is:

- (a) a bijection
- (b) one-one but not onto
- (c) onto but not one-one
- (d) neither one-one nor onto

(iii) Write the principal value of  $\sin^{-1}(-1/2)$ .

(iv) Find the value of  $\tan^{-1}(1) + \cos^{-1}(-1/2)$ .

(v) If  $A =$

|   |   |
|---|---|
| 1 | 2 |
| 3 | 4 |

find  $A + A'$ .

(vi)  $A$  is a square matrix of order 3 satisfying  $A' = -A$ . Write the value of the diagonal element  $a_{11}$ .

(vii) Find the inverse of

|   |   |
|---|---|
| 2 | 0 |
| 0 | 3 |

(viii) If  $A$  is a  $2 \times 2$  matrix with  $|A| = 4$ , find  $|\text{adj } A|$ .

(ix) Is the function

$$f(x) = \begin{cases} (x^2 - 1)/(x - 1) & x \neq 1 \\ 2 & x = 1 \end{cases}$$

continuous at  $x = 1$ ? Justify.

(x) Differentiate  $\tan^{-1}(x^2)$  with respect to  $x$ .

(xi) If  $y = \log(\log x)$ , find  $dy/dx$ .

(xii) If  $x = a \cos \theta$  and  $y = a \sin \theta$ , find  $dy/dx$ .

(xiii) Find the point on the curve  $y = x^2$  at which the tangent is parallel to the  $x$ -axis.

(xiv) The function  $f(x) = e^{2x}$  is strictly increasing on:

- (a)  $\mathbb{R}$
- (b)  $(0, \infty)$  only
- (c)  $(-\infty, 0)$  only
- (d) no interval

(xv) Evaluate  $\int e^{3x} dx$ .

(xvi) Evaluate  $\int_0^{\pi/2} \cos x dx$ .

(xvii) Evaluate  $\int_0^1 1/(1 + x^2) dx$ .

(xviii) Find the area bounded by  $y = x^2$ , the  $x$ -axis and the lines  $x = 0$ ,  $x = 3$ .

(xix) The feasible region of an LPP is unbounded and  $M$  is the largest value of  $Z = ax + by$  at a corner point. Then  $M$  is the maximum value of  $Z$  if:

- (a) the open half plane  $ax + by > M$  has no point in common with the feasible region
- (b) the feasible region is contained in  $ax + by > M$
- (c)  $M$  is attained at two corner points
- (d)  $M > 0$

(xx) **Assertion (A):** if two rows of a determinant are identical, its value is 0.

**Reason (R):** interchanging two rows of a determinant changes its sign.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

## SECTION B - 14 MARKS

### Question 2 [2]

Let  $f : \mathbb{N} \rightarrow \mathbb{N}$  be defined by  $f(n) = n^2 + n + 1$ . Show that  $f$  is one-one but not onto.

### Question 3 [2]

Prove that  $2 \tan^{-1} 1/3 = \tan^{-1} 3/4$ .

OR

Find the value of  $\sin^{-1}(\sin 5\pi/6)$ .

### Question 4 [2]

If  $A =$

$$\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$$

show that  $A^2 - 5A + 7I = O$ .

### Question 5 [2]

The radius of a circle is increasing at  $0.7$  cm/s. Find the rate at which its circumference increases.

### Question 6 [2]

If  $x^2 + y^2 = 4$ , find  $dy/dx$ .

OR

If  $y = (\log x)^{\cos x}$ , find  $dy/dx$ .

**Question 7 [2]**

Evaluate  $\int x e^x dx$ .

**Question 8 [2]**

Find the maximum value of  $Z = 3x + 4y$  subject to  $x + y \leq 4$ ,  $x \geq 0$ ,  $y \geq 0$ .

**OR**

Write the corner points of the feasible region determined by  $x + 2y \leq 8$ ,  $x \geq 0$ ,  $y \geq 0$ .

**SECTION C - 21 MARKS**

**Question 9 [3]**

Show that the relation  $R$  on  $A = \{1, 2, 3, 4, 5\}$  given by  $R = \{(a, b) : |a - b| \text{ is even}\}$  is an equivalence relation. Write its equivalence classes.

**Question 10 [3]**

Using properties of determinants, prove that

$$\begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(4-x)^2$$

**Question 11 [3]**

If  $y = e^{a \cos^{-1} x}$ ,  $-1 \leq x \leq 1$ , show that  $(1 - x^2)(d^2y)/(dx^2) - x(dy/dx) - a^2y = 0$ .

**OR**

Find  $a$  and  $b$  so that

$$\begin{cases} 5 & x \leq 2 \\ ax + b & 2 < x < 10 \\ 21 & x \geq 10 \end{cases}$$

is continuous everywhere.

**Question 12 [3]**

Find the intervals in which  $f(x) = 2x^3 - 9x^2 + 12x + 15$  is (i) strictly increasing, (ii) strictly decreasing.

**OR**

Find the equations of the tangent and the normal to the curve  $y = x^3 - 3x + 2$  at the point  $(2, 4)$ .

**Question 13 [3]**

Evaluate  $\int (x+3)/(\sqrt{5-4x-x^2}) dx$ .

**OR**

Evaluate  $\int (2x+1)/((x+2)(x-3)) dx$ .

**Question 14 [3]**

Evaluate  $\int_{-5}^5 |x+2| dx$ .

**Question 15 [3]**

Solve graphically: minimise  $Z = 3x + 5y$  subject to  $x + 3y \geq 3$ ,  $x + y \geq 2$ ,  $x \geq 0$ ,  $y \geq 0$ .

**SECTION D - 25 MARKS**

**Question 16 [5]**

Prove that  $\tan^{-1}(\sqrt{1+x} - \sqrt{1-x})/(\sqrt{1+x} + \sqrt{1-x}) = \pi/4 - (1/2)\cos^{-1}x$ , for  $-1/\sqrt{2} \leq x \leq 1$ .

**OR**

Show that  $f: \mathbb{R} \rightarrow \{y \in \mathbb{R} : -1 < y < 1\}$  given by  $f(x) = x/(1 + |x|)$  is a bijection.

**Question 17 [5]**

Solve the following system of equations by the matrix method:

$$x + 2y - 3z = -4, \quad 2x + 3y + 2z = 2, \quad 3x - 3y - 4z = 11$$

**Question 18 [5]**

An open box with a square base is to be made out of a given quantity of cardboard of area  $c^2$  square units. Show that the maximum volume of the box is  $c^3/(6\sqrt{3})$  cubic units.

**OR**

A window is in the form of a rectangle surmounted by a semicircular opening, the total perimeter being 10 m. Find the dimensions that maximise the area.

**Question 19 [5]**

Evaluate  $\int \frac{2x}{(x^2 + 1)(x^2 + 3)} dx$ .

**Question 20 [5]**

Find the area of the region bounded by the curve  $y = x^2$ , the line  $y = 4$  and the y-axis, in the first quadrant.

**OR**

Find the area of the region bounded by the parabola  $x^2 = 4y$  and the line  $x = 4y - 2$ .

## ANSWER KEY & MARKING NOTES

Practice Paper 2. **Do not read this section before attempting the paper.** Answers were checked against a computer algebra system; the notes give the decisive step rather than a full solution.

### Section A

| Q         | Answer                                                                                                                                    |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1 (i)     | (a) not reflexive. Take $a = 1/2$ : then $a^2 = 1/4$ and $1/2 \leq 1/4$ is false.                                                         |
| 1 (ii)    | (a) a bijection. $x^3$ is strictly increasing, so one-one; and every real has a real cube root, so onto.                                  |
| 1 (iii)   | $-\pi/6$ . The range of $\sin^{-1}$ is $[-\pi/2, \pi/2]$ and $\sin(-\pi/6) = -1/2$ .                                                      |
| 1 (iv)    | $(11\pi)/12$ . $\tan^{-1}1 = \pi/4$ and $\cos^{-1}(-1/2) = 2\pi/3$ , and $\pi/4 + 2\pi/3 = (11\pi)/12$ .                                  |
| 1 (v)     | $A + A' = [[2, 5], [5, 8]]$ . (The sum of a matrix and its transpose is always symmetric.)                                                |
| 1 (vi)    | 0. Skew symmetry forces $a_{ii} = -a_{ii}$ for every $i$ , so every diagonal entry is 0.                                                  |
| 1 (vii)   | $[[1/2, 0], [0, 1/3]]$ . For a diagonal matrix the inverse is diagonal with the reciprocals.                                              |
| 1 (viii)  | $ \text{adj } A  =  A ^{n-1} = 4^1 = 4$ .                                                                                                 |
| 1 (ix)    | Yes. For $x \neq 1$ the expression is $x + 1$ , whose limit at $x = 1$ is 2, and $f(1) = 2$ .                                             |
| 1 (x)     | $2x/(1 + x^4)$ .                                                                                                                          |
| 1 (xi)    | $1/(x \log x)$ .                                                                                                                          |
| 1 (xii)   | $-\cot \theta$ . $dy/dx = (a \cos \theta)/(-a \sin \theta) = -\cot \theta$ .                                                              |
| 1 (xiii)  | $(0, 0)$ . $dy/dx = 2x = 0$ gives $x = 0$ .                                                                                               |
| 1 (xiv)   | (a) $\mathbb{R}$ . $f(x) = 2e^{2x} > 0$ for every real $x$ .                                                                              |
| 1 (xv)    | $(e^{3x})/3 + C$ .                                                                                                                        |
| 1 (xvi)   | 1. $[\sin x]_0^{\pi/2} = 1 - 0$ .                                                                                                         |
| 1 (xvii)  | $\pi/4$ . $[\tan^{-1}x]_0^1 = \tan^{-1}1 - \tan^{-1}0$ .                                                                                  |
| 1 (xviii) | 9 square units. $\int_0^3 x^2 dx = [x^3/3]_0^3 = 9$ .                                                                                     |
| 1 (xix)   | (a). This is the standard test for an unbounded region; without it the corner point value need not be the maximum.                        |
| 1 (xx)    | (a). Interchanging the two identical rows leaves the determinant unchanged but must also change its sign, so $D = -D$ and hence $D = 0$ . |

### Section B

| Q      | Answer                                                                                                                                                                                                                     |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2      | One-one: for $n \geq 1$ the function is strictly increasing, so $f(n_1) = f(n_2)$ forces $n_1 = n_2$ . Not onto: $f(1) = 3$ is the smallest value, so 1 and 2 have no pre-image.                                           |
| 3      | $2 \tan^{-1}t = \tan^{-1}2t/(1 - t^2)$ with $t = 1/3$ gives $\tan^{-1}(2/3)/(8/9) = \tan^{-1}3/4$ .                                                                                                                        |
| 3 (OR) | $\pi/6$ . $5\pi/6 \notin [-\pi/2, \pi/2]$ , but $\sin 5\pi/6 = \sin(\pi - 5\pi/6) = \sin \pi/6$ .                                                                                                                          |
| 4      | $A^2 = [[8, 5], [-5, 3]]$ , $5A = [[15, 5], [-5, 10]]$ and $7I = [[7, 0], [0, 7]]$ ; $A^2 - 5A + 7I$ is the zero matrix.                                                                                                   |
| 5      | $C = 2\pi r$ , so $dC/dt = 2\pi(0.7) = 1.4\pi$ cm/s ( $\approx 4.40$ cm/s).                                                                                                                                                |
| 6      | Differentiating implicitly, $2x + 2ydy/dx = 0$ , so <b><math>dy/dx = -x/y</math></b> .                                                                                                                                     |
| 6 (OR) | $\log y = \cos x \cdot \log(\log x)$ . Differentiating, $(1/y)(dy/dx) = -\sin x \cdot \log(\log x) + (\cos x)/(x \log x)$ , so <b><math>dy/dx = (\log x)^{\cos x}[-\sin x \log(\log x) + (\cos x)/(x \log x)]</math></b> . |
| 7      | By parts with $u = x$ : $x e^x - \int e^x dx = e^x(x - 1) + C$ .                                                                                                                                                           |
| 8      | Corner points $(0,0)$ , $(4,0)$ , $(0,4)$ give $Z = 0, 12, 16$ . <b>Maximum <math>Z = 16</math> at <math>(0, 4)</math>.</b>                                                                                                |
| 8 (OR) | $(0, 0)$ , $(8, 0)$ and $(0, 4)$ .                                                                                                                                                                                         |

### Section C

| Q       | Answer                                                                                                                                                                                                                                                 |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9       | Reflexive: $ a - a  = 0$ is even. Symmetric: $ a - b  =  b - a $ . Transitive: if $a - b$ and $b - c$ are both even so is their sum $a - c$ . Two numbers are related exactly when they have the same parity.<br><b>Classes: {1, 3, 5} and {2, 4}.</b> |
| 10      | Apply $C_1 \rightarrow C_1 + C_2 + C_3$ : every entry of the first column becomes $5x + 4$ . Take it out, then use $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$ to obtain two factors of $(4 - x)$ . Verified symbolically.             |
| 11      | $dy/dx = (-ay)/(\sqrt{1 - x^2})$ , so $\sqrt{1 - x^2} dy/dx = -ay$ . Squaring and differentiating once more removes the radical and gives the result. Verified symbolically.                                                                           |
| 11 (OR) | Continuity at $x = 2$ : $2a + b = 5$ . At $x = 10$ : $10a + b = 21$ . Subtracting, $8a = 16$ , so <b><math>a = 2</math> and <math>b = 1</math></b> .                                                                                                   |
| 12      | $f(x) = 6x^2 - 18x + 12 = 6(x - 1)(x - 2)$ . <b>Increasing on <math>(-\infty, 1)</math> and <math>(2, \infty)</math>; decreasing on <math>(1, 2)</math>.</b>                                                                                           |
| 12 (OR) | $dy/dx = 3x^2 - 3 = 9$ at $x = 2$ , and the point does lie on the curve ( $8 - 6 + 2 = 4$ ). <b>Tangent: <math>y = 9x - 14</math>. Normal: <math>x + 9y = 38</math></b> (slope $-1/9$ ).                                                               |
| 13      | Write $x + 3 = -1/2(-4 - 2x) + 1$ and complete the square, $5 - 4x - x^2 = 9 - (x + 2)^2$ . The two parts give <b><math>-\sqrt{5 - 4x - x^2} + \sin^{-1}((x + 2)/3) + C</math></b> .                                                                   |
| 13 (OR) | Partial fractions give $(3/5)/(x + 2) + (7/5)/(x - 3)$ , so the integral is <b><math>3/5 \log x + 2  + 7/5 \log x - 3  + C</math></b> .                                                                                                                |
| 14      | Split at $x = -2$ . $\int_{-5}^{-2} -(x + 2) dx = 9/2$ and $\int_{-2}^5 (x + 2) dx = 49/2$ , so the value is <b>29</b> .                                                                                                                               |
| 15      | The region is unbounded. Corner points $(3, 0)$ , $(3/2, 1/2)$ and $(0, 2)$ give $Z = 9, 7$ and $10$ . Since $3x + 5y < 7$ has no point in common with the region, <b>minimum <math>Z = 7</math> at <math>(3/2, 1/2)</math></b> .                      |

## Section D

| Q       | Answer                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16      | Put $x = \cos 2\theta$ , so $\theta = 1/2 \cos^{-1}x$ . Then $\sqrt{1 + x} = \sqrt{2} \cos \theta$ and $\sqrt{1 - x} = \sqrt{2} \sin \theta$ , and the argument becomes $(\cos \theta - \sin \theta)/(\cos \theta + \sin \theta) = \tan(\pi/4 - \theta)$ . Hence the value is $\pi/4 - \theta = \pi/4 - (1/2)\cos^{-1}x$ .                                                                                                           |
| 16 (OR) | For $x \geq 0$ , $f(x) = x/(1 + x) \in [0, 1)$ ; for $x < 0$ , $f(x) = x/(1 - x) \in (-1, 0)$ . $f$ is strictly increasing on each piece and across 0, hence one-one, and its range is exactly $(-1, 1)$ . Inverse: $f^{-1}(y) = y/(1 -  y )$ .                                                                                                                                                                                      |
| 17      | $A = [[1, 2, -3], [2, 3, 2], [3, -3, -4]]$ has $ A  = 67 \neq 0$ , so $X = A^{-1}B$ gives <b><math>x = 3, y = -2, z = 1</math></b> . Check: $3 - 4 - 3 = -4 \checkmark$ , $6 - 6 + 2 = 2 \checkmark$ , $9 + 6 - 4 = 11 \checkmark$ .                                                                                                                                                                                                 |
| 18      | Let the base be $x \times x$ and the height $h$ . Then $x^2 + 4xh = c^2$ , so $h = (c^2 - x^2)/4x$ and $V = x^2h = (x(c^2 - x^2))/4$ . $dV/dx = (c^2 - 3x^2)/4 = 0$ gives $x = c/\sqrt{3}$ , and $V$ then equals $(\sqrt{3}c^3)/18 = c^3/(6\sqrt{3})$ . The second derivative is negative there.                                                                                                                                     |
| 18 (OR) | With semicircle radius $r$ and rectangle height $h$ , $2r + 2h + \pi r = 10$ and $A = 10r - 2r^2 - (\pi r^2)/2$ . Setting $dA/dr = 0$ gives $r = 10/(\pi + 4)$ and then $h = r$ . <b>Radius = height = <math>10/(\pi + 4)</math> m <math>\approx 1.40</math> m, width = <math>20/(\pi + 4)</math> m <math>\approx 2.80</math> m, maximum area = <math>50/(\pi + 4)</math> m<sup>2</sup> <math>\approx 7.00</math> m<sup>2</sup>.</b> |
| 19      | Put $t = x^2$ , so $dt = 2x dx$ and the integral becomes $\int dt/((t + 1)(t + 3))$ . Partial fractions give $1/2[1/(t + 1) - 1/(t + 3)]$ , so the value is <b><math>1/2 \log(x^2 + 1)/(x^2 + 3) + C</math></b> .                                                                                                                                                                                                                    |
| 20      | Integrating along the $y$ -axis, area = $\int_0^4 \sqrt{y} dy = [2/3 y^{3/2}]_0^4 = 16/3$ <b>square units</b> .                                                                                                                                                                                                                                                                                                                      |
| 20 (OR) | The curves meet where $x^2 = x + 2$ , i.e. $x = -1$ and $x = 2$ . Area = $\int_{-1}^2 [(x + 2)/4 - x^2/4] dx = 9/8$ <b>square units</b> .                                                                                                                                                                                                                                                                                            |